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# Recent Advances in Bio-Based Sustainable Aliphatic and Aromatic Epoxy Resins for Composite Applications

 1167

## Abstract:

In few recent years, it is seen that there is a rapid expansion in the area of bio based thermosetting resins sighting the reason that it has bright future. The curiosity in developing bio based products and bio based refinery processes has been strengthened due to the swift exhaustion of petroleum and also due to new environmental set of laws. The mostly used epoxy resins are by and large diglycidylether of bisphenol A (DGEBA), which is petroleum based ones owing to its superior mechanical and thermal properties. But the research is in full fledge to hit upon a bio-based sustainable substitute for DGEBA. As the researches over the years have shown that, bio-epoxy derived from neither natural oil nor lignin derivatives or other bio-based aromatic resins are able to

fully replace DGEBA in terms of all properties. Hence, it opens a wide window open for their blends with DGEBA. This paper will shower a light on the current progresses made in the field of bio-based epoxy monomers derived from different natural oils and its blends with DGEBA, lignin derived aromatic resins and finally a novel bio-epoxy derived from magnolol. The magnolol derived diglycidyl ether of magnolol (DGEM) was found to have comparable mechanical and thermal properties with better flame retardancy and hence could be a good contender to replace DGEBA.



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